Unit Final Assessment

What Have I Learned About Energy?

Grade 4 Energy Transfer & Transformation · 26 items · 50 points

Name: _____ Date: _____

Directions: Read each item carefully. Show your evidence. You may use your Core Vocabulary cards.

Section 1 — Energy Causes Change

1. Which show evidence of energy causing a change? Circle ALL correct answers.

a) A basketball hits the backboard.

b) You get a small shock from a doorknob.

c) The school bus engine makes a sound when it starts.

d) A candle flame lights up a dark room.

2. Energy is the ability to _____.

a) stay still

b) cause change

c) make heat only

d) be invisible

3. Draw a model of energy causing a change. Show a BEFORE and an AFTER. Label the type of energy. Write one sentence describing the change.

BEFORE → AFTER

Type of energy: _____

Describe the change: _____

4. Explain how an observable change is EVIDENCE that energy is involved.

Section 2 — Moving Objects Have Energy

5. A moving object has _____.

a) stored energy only

b) no energy

c) energy of motion

d) only sound energy

6. Which shows a moving object causing a change?

a) A ball sitting on a shelf

b) A rolling ball knocks over a block

c) A pencil lying on a desk

d) A book held still in your hand

7. Give ONE example of a moving object causing a change. Describe the change.

Section 3 — Speed and Energy (Same Object)

8. Read the table. The SAME toy car rolled down a ramp at three heights and pushed a cup.

Ramp height	Cup movement (measured)	Energy of motion (inferred)
Low	moved a little	less
Medium	moved more	more
High	moved the most	most

What pattern do you see?

9. In the ramp test, what do we MEASURE, and what do we INFER?

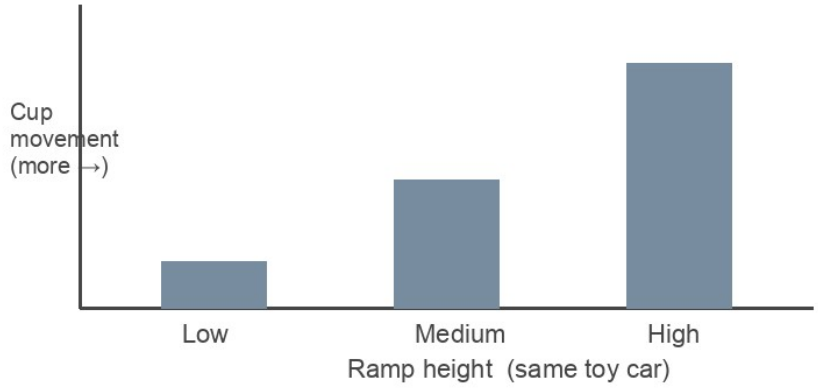
a) measure cup movement, infer speed

b) measure speed, infer cup movement

c) measure energy, infer height

d) measure height, infer the car's color

10. Look at the graph. It shows how far the cup moved at different ramp heights for the SAME car.



As ramp height increases, cup movement _____. This is evidence that a faster car has _____ energy of motion.

11. For the SAME object, what is the relationship between speed and energy of motion? Use evidence.

Section 4 — Energy Transfer

12. Which shows sound energy transferring from place to place?

a) A bell rings and you hear it across the room
b) A ball sits still on the ground
c) A block stays on a table
d) A ruler rests in a drawer

13. Warm water warms a cool spoon. The heat MOVES but stays heat. This is _____.

a) TRANSFER
b) TRANSFORMATION
c) BOTH
d) neither

14. Which are evidence that energy TRANSFERRED from place to place? Circle ALL correct.

a) You hear a bell from far away
b) A book stays closed on a desk
c) You flip a switch and the room lights up
d) A pencil does not move
e) A backpack left in the sun gets warm

15. A flashlight turns on. Stored energy in the battery becomes light. This is _____.

a) TRANSFER
b) TRANSFORMATION
c) neither

16. Label the energy change in a flashlight:

INPUT: _____ → OUTPUT: _____

Section 5 — Collisions

17. During a collision, what happens to the energy of motion?

a) It disappears
b) It transfers to the target, and some becomes sound
c) It becomes cold only
d) Nothing happens

18. Read the table: a toy car rolls into a stack of blocks.

Car	Weight changed?	Blocks knocked over
Test 1	no extra weight	1 block
Test 2	added weight (ONLY change)	2 blocks

Explain the result using energy:

19. Is the test in item 18 a fair test? Why or why not? Predict what happens if you add even more weight.

20. After a collision, the energy _____.

a) is gone forever
b) turns into light only
c) does not disappear — it transfers, and some becomes sound
d) becomes nothing

Section 6 — Transformations and Engineering

21. A domino falls and clicks. Motion passes to the next domino AND some becomes sound. This is _____.

a) TRANSFER only
b) TRANSFORMATION only
c) BOTH
d) neither

22. Draw a line to match each word to its meaning.

criteria — the limits (materials, time, safety)
constraints — what a good solution must do
refine — to improve using test evidence

23. Pick ONE device (flashlight, buzzer, or ramp toy). Write a claim about its energy change, give evidence, and explain your reasoning. Use "supports," not "proves."

Claim: _____

Evidence: _____

Reasoning: _____

24. Label the energy chain for a fan:

INPUT: electrical energy → TRANSFORMATION: _____ → OUTPUT: the fan spins

Section 7 — Edison and Engineering Persistence

25. What is TRUE about Thomas Edison and the light bulb?

a) He invented the first light bulb all by himself

b) He improved earlier designs by testing and redesigning

c) He never tested any materials

d) He gave up after one try

26. How did Edison use TESTING and REDESIGN to solve problems? What does this show about persistence?

End of assessment.

Provenance: ADAPTED — original assessment aligned to the official Unit Assessment blueprint (50 pts), re-authored.